

JOCUND MISSIER

Senior Software Engineer – Automotive Embedded & AUTOSAR

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PROFESSIONAL SUMMARY

Analytical Senior Software Engineer with 5 years in Classic & Adaptive AUTOSAR, ADAS, and embedded systems. Proficient in C/C++, QNX, Linux/Yocto, ROS, and hardware platforms including NXP TDA4 and Raspberry Pi 5. Experienced in MISRA compliance, HIL testing, CAN/SPI/I2C protocol implementation, and ASPICE/Agile delivery across leading automotive OEM programs.

TECHNICAL SKILLS

- **Languages:** C, C++, Python, Linux Shell Scripting, ROS Script
- **Protocols:** CAN, LIN, UART, I2C, SPI | **Networking:** TCP/IP, Socket Programming
- **OS & Platforms:** Linux (Ubuntu / Yocto), QNX, VxWorks | **HW:** NXP TDA4, RPI3 / RPI5, Ti-Tda4
- **AUTOSAR / Standards:** Classic & Adaptive AUTOSAR, ASPICE, MISRA, Agile, SDLC, ISO 26262
- **Tools:** VectorCAST, Canoe, GCov, Axivion, CMake, Bazel, Conan, GDB, EA v12, Simics (Virtual Platform), Polarion (Requirements & SAD), Jenkins (CI/CD)
- **Testing:** SWIT Test Planning & Execution, HIL Testing, Unit Testing, VIP Image Validation
- **IDEs & VCS:** VSCode, Eclipse, QNX Momentics, PyCharm | Git, Gerrit, Bitbucket (Branch Merge & CCR Approvals), Tortoise SVN

PROFESSIONAL EXPERIENCE

Senior Software Engineer – Tata Elxsi (Client: WindRiver)

Aug 2025 - Present

Kernel & Safety Diagnostics Development and testing | TDA4 HW | VxWorks | C

- Developed VxWorks kernel-space device drivers for INTA, UDMA, and Pinmux modules targeting TI TDA4 automotive SoC
- Implemented Device Tree Source (DTS) configurations for hardware peripheral binding, ensuring correct resource allocation and interrupt routing
- Designed INTA interrupt handling routines with ISR registration and priority configuration in VxWorks RTOS; implemented UDMA channel init and DMA descriptor management for safety-critical data paths
- Analysed and traced customer requirements from Polarion (SAD documentation) to ensure traceability across design, implementation, and SWIT test cases
- Authored and reviewed Software Interface Test (SWIT) plans and test reports; executed validation on both Simics virtual platform and physical TDA4 hardware
- Performed code reviews for team members and jira tracking; provided technical guidance on kernel-space driver implementation and user-space test harness development
- Managed code integration lifecycle — raised and approved Change Control Records (CCR), merged branches in Bitbucket, and monitored CI/CD pipeline status via Jenkins
- Validated VIP image files on Simics and HW targets; performed end-to-end debugging across kernel and user space to isolate and resolve peripheral initialisation failures
- Collaborated with hardware team using Simics simulator for peripheral debug; maintained traceability matrices aligned with ISO 26262 guidelines

Senior Software Engineer – Tata Elxsi (Client: Aptiv)

Feb 2025 – July 2025

CPU Load & Performance | TDA4 HW | RTOS/ BareMetal/ Win-Linux | C++14

- Developed CPU load monitoring and profiling framework for TI TDA4 SoC targeting the Cortex-A72 application core in both RTOS and Linux environments.
- Implemented C++ based load measurement utilities leveraging TI Processor SDK (PSDK) APIs for task-level CPU utilization tracking.
- Validated CPU load measurements against hardware performance counters to ensure accuracy of profiling results under varying workload conditions.

Senior Software Engineer – Tata Elxsi (Client: Aptiv)

July 2023 – Dec 2024

ADAS Driving & Parking Module | NXP TDA4 HW | QNX / VxWorks / Win-Linux | C++14

- Developed ADAS Parking Stack on SW components; executed HIL testing across multiple OS platforms
- Validated unit tests and fixed Coverity violations per MISRA guidelines using Axivion
- Implemented ROS integration & StradVision parser for ADAS perception pipeline
- Built Yocto image for customised TiTda4 board enabling cross-compilation and OS benchmarking

Software Engineer – Tata Elxsi (Client: JLR)

Nov 2023 – Feb 2024

Freshness Value Manager (FVM) | Adaptive AUTOSAR | ARA COM Functional Cluster

- Created comprehensive test cases for FVM stack; executed Unit Testing with VCast on Windows & Linux
- Resolved MISRA Coverity violations, ensuring security for diagnostics PDU data handling

Software Engineer – Avin Systems (Client: ZF Friedrichshafen)

July 2021 – May 2023

Basis SW Platform for ECUs | Adaptive AUTOSAR | POSIX OS | QNX / Linux

- Built basis software platform for performance ECUs with service-oriented architecture on POSIX OS
- Set up DevOps pipeline (Jumpstart Accelerator); configured pre-integrated SW stack with Adaptive AUTOSAR
- Standardised ZF APIs across OTA, IOT, SiLetc on QNX and Linux; implemented CAN/SPI/I2C middleware

PERSONAL PROJECTS – RASPBERRY PI 5

CAN Bus Monitor & Signal Decoder

RPI5 · MCP2515 · Python / C

- Interfaced MCP2515 CAN controller via SPI; captured live frames and decoded signals using a custom DBC parser
- Logged and visualised CAN traffic with Python – directly mirrors automotive diagnostic workflows

ROS2 Sensor Fusion Node (Obstacle Detection)

RPI5 · ROS2 Humble · IMU / Ultrasonic

- Deployed ROS2 Humble on RPI5; built publisher/subscriber nodes for multi-sensor fusion pipeline
- Applied ADAS domain knowledge for real-time obstacle detection and data visualisation in RViz

I2C Multi-Sensor Dashboard

RPI5 · C / Python · GPIO

- Connected multiple I2C devices (temperature, pressure, accelerometer); wrote low-level register access in C
- Validated signal integrity with oscilloscope; built real-time terminal dashboard in Python

EDUCATION & CERTIFICATIONS

B.E. Computer Science & Engineering

2020

Dr. Sivanthi Aditanar College of Engineering, Tamil Nadu

IoT & Data Science Training – Cranes Varsity, Bangalore

2021

Certifications:

- Cisco CCNA 200-301 – Udemy (April 2021)
- Image Processing & Computing Programming – GUVI (April 2021)